

## UNIT TWO

### REVIEW OF MATRIX ALGEBRA AND ITS APPLICATION

#### Introduction

Numerical data arranged in a form that we shall come to call a matrix are very common in everyday life. Most linear equations of any consequence involve large numbers of equations and variables (unknowns). Computer is heavily utilized to solve such linear equations. However, there are two activities that computers cannot perform for you: formulation problems in terms of linear systems and interpreting the results. Therefore, though computers help you to solve problems, no one can deny the importance of hand methods.

In the upcoming sections, we will try to cover some concepts about matrix algebra, inverse of matrices, solving system of linear equations (Using Gauss - Jordan and inverse methods), which have many importance in solving the real world problems like input - output model, Nutrition, income tax, life sciences, etc and Markov chains which are heavily utilized in the analysis of trend of a market.

#### Review on matrix algebra

Meaning of Matrix

What is matrix? Can you define in your own words?

Matrix could be defined in different ways having the central or core idea unchanged. These differences simply mount to wording. Some of the definitions are:

A matrix is a rectangular array of numbers. Alternatively, matrix is a rectangular array of numbers arranged in rows and columns. In other words, it is a collection of numbers which are arranged in rows and columns. The word "Matrices" is the plural form of matrix. Numbers which form matrices are enclosed by using symbols like parentheses, ( ) or brackets [ ]. Higher case letters are used to represent matrix and small letters are used to represent numbers which form a matrix. Each member in a matrix is called an **element**.

#### Example 2.1

$$A = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

elements

In general, matrix A containing elements  $a_{ij}$  has the general form:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix} \text{ mxn, where}$$

in  $a_{ij}$ , i represents the row

j represents the column

mxn represents dimension of matrix, where

m is number of rows and

n is number of columns.

In the above matrix, A,  $a_{12}$  is read as "a sub one two" and  $a_{12}$  represents the element (entry) of the matrix and subscript "12" represents the position of the entry. The entry  $a_{12}$  is located at the intersection of row one

and column 2. Row represents the horizontal array of numbers and column represents the vertical array of numbers.

Dimension of a matrix represents the order, shape or size of a matrix. Since matrix "B" above has two rows and two columns, its dimension is 2x2 (read as two by two) and is called a two by two matrix.

? What do you think  $a_{23}$ ,  $a_{31}$  and the subscripts represents? (Use, the space below)

### Example 2.2

These illustrative examples will help you to clarify any ambiguity.

1.  $[0]$  is a 1x1 matrix, because it has only one row and one column.

2.  $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$  is a 1x3 matrix.

3.  $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$  is a 2x1 matrix.

4.  $\begin{pmatrix} 0 & 2 & 4 \\ 2 & 1 & 0 \end{pmatrix}$  is a 2x3 matrix.

5.  $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 0 & 2 & 4 \end{pmatrix}$  is a 4x3 matrix.

 **Note1:** To avoid having to specify which of a pair of subscripts represent the row and, which the column, we shall always use ‘the row, column’ order convention, where the 1<sup>st</sup> subscript “1” in  $a_{12}$  for example refers to row and the second subscript “2” refers to column.

**2:** A comma is required to avoid ambiguity if the number of rows or columns exceeds nine. For example, an entry in the 12<sup>th</sup> row and 3<sup>rd</sup> column would be written as  $a_{12, 3}$ .

### Example 2.3

$$A = \begin{pmatrix} 75 & 82 & 86 \\ 65 & 70 & 68 \\ 73 & 76 & 74 \\ 59 & 80 & 99 \end{pmatrix}$$

In the above matrix A, determine:

i) The dimension.

ii)  $a_{22} =$  \_\_\_\_\_

iii)  $a_{31} =$  \_\_\_\_\_

iv)  $a_{43} =$  \_\_\_\_\_

v)  $a_{55} =$  \_\_\_\_\_

### Types of Matrices

Dear learners, though there are many matrices, we will try to discuss only some of them.

a) Vector matrix- This is a matrix which has precisely one row or precisely one column. From this definition, we can identify two classifications.

i) Row matrix is a matrix having only one row. It is a 1xn matrix, and is called a row vector of dimension n, i.e., 1xn matrix.

#### Example 2.4

$A = [2 \quad 1 \quad 4]$  is a 1x3 matrix.

$B = \begin{bmatrix} 1 & 2 \end{bmatrix}$  is a  $1 \times 2$  matrix.

- i) **Column matrix**- is a matrix having only one column. It is an  $m$  by  $1$  matrix, and is also called a column vector of dimension  $m$ , i.e.,  $m \times 1$  matrix.

### Example 2.5

$$C = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad D = \begin{pmatrix} 4 \\ 9 \end{pmatrix} \quad \begin{matrix} 2 \times 1 \text{ matrix} \\ 3 \times 1 \text{ matrix} \end{matrix}$$

 **Note:** Entries in row and column vectors often are referred to as components of the vectors.

- b) **Square matrix** is a matrix having the same number of rows and columns, i.e., for two or more matrices to be square matrices, they should have to have the same shape or size. If the dimension of a matrix is  $m \times n$ , for the square matrix,  $m = n$ ,  $m \times n = m \times m = n \times n$ .

### Example 2.6

$$C = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 2 \\ 4 & 3 & 7 \end{pmatrix} \text{ is a } 3 \times 3 \text{ matrix.}$$

$$D = \begin{pmatrix} 1 & 0 \\ 4 & 3 \end{pmatrix} \text{ is a } 2 \times 2 \text{ matrix.}$$

- c) **Equal matrices** – For two or more matrices to be said equal matrices, necessarily they should have the same shape and the same corresponding entries.

$$G = \begin{pmatrix} 2 & 3 & 5 \\ 5 & 0 & 6 \\ 1 & 2 & 1 \end{pmatrix} \text{ is a } 3 \times 3 \text{ matrix.} \quad H = \begin{pmatrix} 2 & 6/2 & 6-1 \\ -1+6 & 0 & 6 \\ 1 & 4/2 & 2-1 \end{pmatrix}$$

Matrices  $G$  and  $H$  are equal matrices, because the two are  $3 \times 3$  matrices and their corresponding entries are the same.

- d) **Identity Matrix or Unit Matrix** is a square matrix where elements along the primary or main diagonal all equal to one (1), while all other elements are equal to zero. Main diagonal is a line that runs from upper left to lower right. Identity matrix is denoted by  $I$ .

### Example 2.7

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ is a } 3 \times 3 \text{ matrix.} \quad \text{Main diagonal} \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} 2 \times 2$$

$$C = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \text{ is a } 3 \times 3 \text{ matrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} 2 \times 3$$

In the above example, matrix A and B are identity matrices, because they are both square matrices and elements on the main diagonal are 1 and all other elements are zero. But C and D are not square matrices. Can you mention the reasons?

(e) *Zero Matrix* is a matrix whose elements are all zeros and is designated by O.

**Example 2.8**

$$V = 0 \begin{pmatrix} 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The above matrices are zero matrices.

f) *Scalar Matrix* is a matrix with a constant number on its main diagonal and zeros in all other locations.

**Example 2.9**

$$A = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \quad B = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

The above matrices are scalar matrices.

g) *Diagonal Matrix* is a matrix with value other than zero in its main diagonal and all other elements equal to zero.

**Example 2.10**

$$C = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

 **Note:** From the above discussion, we can infer that:

- identity matrix can be both scalar and diagonal matrices
- Identity, scalar and diagonal matrices are square matrices, but the vice versa may not necessarily be true.
- Scalar matrix can be diagonal, but the vice versa may not necessarily be true.

**Activity**

1. Which of the following matrices are:

- i) equal \_\_\_\_\_
- ii) square \_\_\_\_\_
- iii) identity \_\_\_\_\_
- iv) scalar \_\_\_\_\_
- v) diagonal \_\_\_\_\_
- vi) zero \_\_\_\_\_

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 2 & 5 \\ 1 & 3 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 4-2 & 3 & 6-2 \\ 8/2 & 2 & 5 \\ 1 & 6/2 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$E = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad F = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \quad G = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad H = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

## Matrix operations and their properties on matrices

### Addition and Subtraction on Matrices

Matrices are added or subtracted by adding or subtracting the corresponding entries. Since entries must correspond, matrices must involve the same dimensions. Therefore, the sum or difference of two or more matrices of the same size can be obtained by adding or subtracting the corresponding elements of two or more matrices. This is possible only if the matrices have the same size.

#### Example 2.11

$$\text{Let } A = \begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 4 \end{pmatrix} \quad C = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$$

Then,

a)  $A \pm B$

b)  $A \pm C$

What is the sum and difference of the following two matrices?

$$H = \begin{pmatrix} \frac{1}{2} & 6 \\ 4 & 5/2 \\ 0 & 1 \end{pmatrix} \quad J = \begin{pmatrix} 4 & 3 \\ 2/5 & 6 \end{pmatrix}$$

### Properties of Addition and subtraction on matrices

Because we add two matrices by adding their corresponding elements, it follows from the properties of real numbers that matrices of the same dimension are commutative and associative relative to addition.

#### i) Commutative property

For two matrices with the same size, if interchanging of the positions of matrices has no change on the sum, then addition is communicative. Thus, for two matrices A and B,  $A+B = B+A$ .

#### Example 2.13

$$A = \begin{pmatrix} 2 & 4 \\ 3 & 5 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 6 \\ 1 & 2 \end{pmatrix}$$

$$A+B = \begin{pmatrix} 0 & 10 \\ 4 & 7 \end{pmatrix} = B+A$$

But,  $A-B \neq B-A$ , i.e., subtraction is not commutative over matrices.

#### ii) Associative Property

For three matrices, A, B, C with the same dimension,  $(A+B) + C = A + (B+C)$  is called associative property.

#### iii) Identity Property

For any matrix A, there is a corresponding matrix  $-A$ , such that

$A + (-A) = 0 = (-A) + A$ . The negative of a matrix A, denoted by  $-A$ , is a matrix

with elements that are the negative of elements in A. Thus, if

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then, } -A = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}$$

 **Note:**  $A + (-A) = 0$  (a zero matrix) which is an additive identity. If A and B are matrices of the same dimension, then we define subtraction as follows:

$A - B = A + (-B)$ . Thus, to subtract matrix B from matrix A, we simply add the negative of B to A.

#### Example 2.14

$$\begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} - \begin{pmatrix} -2 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} + \begin{pmatrix} 2 & -4 \\ -2 & -3 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -4 & 1 \end{pmatrix}$$

## Activity

1. Find a solution for the following matrices in the spaces provided below.

$$A = \begin{pmatrix} 5 & 8 \\ -1 & 3 \end{pmatrix} \quad B = 2 \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$$

- i)  $A+B$   
ii)  $A-B$   
iii) Compare  $A+B$  with  $B+A$  and  $A-B$  with  $B-A$  make conclusion.
2. Compute  $A+(B+C)$  and  $(A+B)+C$  for

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 5 & -6 \\ 2 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 7 & 1 \\ 8 & 9 \end{pmatrix}$$

3. Find  $w, x, y, z$ , so that

$$\begin{pmatrix} 4 & -2 \\ -3 & 0 \end{pmatrix} + \begin{pmatrix} w & x \\ y & z \end{pmatrix} = \begin{pmatrix} 2 & -3 \\ 0 & 5 \end{pmatrix}$$

## Multiplication of Matrices

Dear students, be informed that the portions which we cover in the following sections are very important to easily cope up with the succeeding unit. Therefore, please pay your attention for the discussions.

Matrix multiplication is a specialized form of multiplication devised for very particular reasons.

## Scalar Multiplication

In matrix algebra, an ordinary number is called scalar. The product between a real number (scalar) and a matrix is called scalar product. To multiply a matrix by a scalar, each entry of a matrix is multiplied by a scalar. Therefore, the product of a number  $K$  and a matrix  $M$ , denoted by  $KM$ , is a matrix formed by multiplying each element of  $M$  by  $K$ .

### Example 2.15

$$B = 2 \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \quad C = -1/2 \begin{pmatrix} 3 & -1 & 0 \\ -2 & 0 & 3 \end{pmatrix} = \begin{pmatrix} -3/2 & 1/2 & 0 \\ 1 & 0 & -3/2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 4 \\ -6 & 8 \end{pmatrix}$$

## Inner Product

Inner product, which is also called *dot product*, refers to the multiplication of elements in row vector with column vector. The result of inner product is always a scalar. Inner product is defined when the number of row vector elements is equal to number of elements of column vector.

Example

Let  $A = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \rightarrow$  a row matrix

$$B = \begin{pmatrix} 5 \\ 2 \\ 6 \end{pmatrix} \rightarrow \text{a column vector}$$

What is the inner product of  $A$  and  $B$ ?

## Matrix Product

The product of two matrices  $A$  and  $B$ ,  $AB$  is defined if and only if the number of rows of matrix  $B$  is the same as number of columns of  $A$ . If this is satisfied, the two matrices are said to be conformable for

multiplication. The dimension of the resulting matrix, AB is the same as number of rows of matrix A and number of columns of matrix B.

Therefore, given matrices A and B having dimension of

$$A(m_A \times n_A) \text{ and } B(m_B \times n_B),$$

$$A.B = C \text{ (new matrix).}$$

$$(m_A \times n_A)(m_B \times n_B) = C(m_A \times n_B)$$

$\Rightarrow$  If  $n_A = m_B$ , the matrices are conformable for multiplication.

Generally, if A is  $a \times b$  matrix, and B is  $c \times d$  matrix, then AB is  $a \times d$  matrix.

Dear students, the definition is not as complicated as it might first seem. An example should help you to clarify the process.

### Example 2.17

For  $A = \begin{pmatrix} 2 & 2 \\ -4 & 1 \end{pmatrix}$  and  $B = \begin{pmatrix} 2 & 3 & 1 \\ -4 & 1 & 8 \end{pmatrix}$ ,

$$A.B = ?$$

Compute the following.

$$A = \begin{pmatrix} 2 & 3 & -1 \\ -2 & 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 3 \\ 2 & 0 \\ -1 & 2 \end{pmatrix} \quad C = \begin{pmatrix} 1 & \frac{1}{2} \\ 4 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & -2 \\ 1 & 4 \\ 2 & 1 \end{pmatrix} \text{ then,}$$

$$AB = \underline{\hspace{2cm}}$$

$$CD = \underline{\hspace{2cm}}$$

### Properties of Multiplication

- Scalar multiplication is distributive and associative with respect to addition.

$$K(A+B) = KA+KB, \text{ where } K \text{ is scalar.}$$

Let  $K=2$ ,  $A = \begin{pmatrix} 4 & -5 \\ 0 & 1 \end{pmatrix}$   $B = \begin{pmatrix} 2 & -4 \\ 0 & 4 \end{pmatrix}$ , then find  $K(A+B)$

- Scalar multiplication also holds true for associative property.

$$K.(A.B) = (K.A).B$$

- Matrix multiplication is **not** commutative, i.e.,  $A.B \neq B.A$ . But, under exceptional conditions it may be commutative:

i) When the two matrices are equal, i.e.,  $A=B$ , hence  $A.B=B.A$ .

ii) When a matrix is multiplied by identity matrix,  $AI=IA$ .

- Matrix multiplication is associative. i.e.,  $A.(B.C) = (A.B).C$ . But the three matrices should be conformable.
- Matrix multiplication by an identity matrix results in original matrix.

**Note: 1.** If  $AB=0$ , this does not imply that  $A=0$ , or  $B=0$  or both A and B are zero matrices. Thus, the zero property does not hold for matrix multiplication.

**2.** If product of AB is defined, this does not necessarily imply BA is defined.

**3.** If  $AB=BC$ , this does not imply  $A=C$ , or if  $AC=BC$ , this does not imply  $A=B$ .

### Example 2.19

$$A = \begin{pmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Generally, assuming all products and sums are defined for the indicated matrices A, B and C, and for K real number:

|                                |                         |
|--------------------------------|-------------------------|
| ✓ Associative Property         | $A(BC) = (AB)C$         |
| ✓ Left distributive property   | $A(B + C) = AB + AC$    |
| ✓ Right Distributive property  | $(B + C)A = BA + CA$    |
| ✓ Left multiplication property | If $A=B$ , then $CA=CB$ |
| ✓ Right multiplication         | If $A=B$ , then $AC=BC$ |
| ✓ Associative property         | $K(AB) = (KA)B = A(KB)$ |

### Activity

Below are question you need to work on. Write your answers in the spaces provided.

1. Given matrices

$$A = \begin{pmatrix} -1 & 1 \\ 2 & 3 \\ 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} -1 & 0 & 3 & -2 \\ 1 & 2 & 2 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{pmatrix}$$

Find

i)  $A B$

ii)  $B A$

2. Compute

$$0.05 \begin{pmatrix} 430 & 212 \\ 210 & 165 \\ 435 & 315 \end{pmatrix} + 0.07 \begin{pmatrix} 234 & 436 \\ 160 & 212 \\ 410 & 136 \end{pmatrix}$$

### Matrix Application

#### Solving System of Linear Equation Using Matrix

An  $m \times n$  system of equations having the forming:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_n, \text{ can be written in a matrix form as:}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

↓
↓
↓

Coefficient of equations
Variables
constant terms

In general, a linear equation of the form  $a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b_1$  can be written in a matrix form as:

$$\begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_n \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = b_1 \text{ this form is called compact form.}$$

#### Example 2.20

$$2x_1 + 5x_2 = 9$$

$$3x_1 + x_2 = 3$$

$-2x_1 + x_2 = 1$ , then the matrix form of this system of linear equation is:

$$\begin{pmatrix} 2 & 5 \\ 3 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 1 \end{pmatrix}$$

### Solving System of Linear Equations

Dear students, now we are moving to the section that discusses about how to solve a system of linear equations which are largely applied in the business problems. Pay your attention here. Linear equations can be solved using different methods, including *Inverse Method*, *Gauss-Jordan Method*, *Cramer's (Determinant's) Rule*, *Simultaneous methods and others*. In this course we focus on the first two methods.

#### Inverse Method

To use this approach, it is necessary to have clear understanding about row operations and inverse of a matrix.

##### i) Elementary Row Operations

Elementary matrix row operations are fundamental when we compute the inverse of matrix, solving system of linear equations and linear programming problems and etc. Therefore, having clear understanding will help you to solve problems easily in the succeeding portions.

There are three elementary matrix row operations. These include:

1) Multiply or Divide a row by a non- zero constant. In this operation, elements of a row are multiplied by a non-zero constant (K) and replaces the original row elements by the resulting elements. i.e.,  $KR_i \rightarrow R_i$ , where,  $i$  represent the  $i_{th}$  row,  $K$  represents a non-zero constant and “ $\rightarrow$ ” stands for “replaces”.

#### Example 2.21

Let  $K=2$  and  $A = \begin{pmatrix} 3 & 4 \\ 2 & 9 \\ 1 & 1 \end{pmatrix}$  then,

i)  $KR_1$ ?

ii)  $1/KR_3$ ?

2) Multiply or dividing a row by a non -zero constant and adding or subtracting the result to or from another row, i.e.,  $KR_i \pm R_j \rightarrow R_j$ , where  $R_i$  represents the row to be multiplied or divided by constant  $K$  and  $R_j$  represents the row to be replaced.

#### Example 2.21

Perform the second elementary row operation by adding or subtracting 2 times elements of second row to or from third row's elements?

$$A = \begin{pmatrix} 5 & 2 & 9 \\ 3 & 1 & 2 \\ 1 & 4 & 6 \end{pmatrix} \quad \text{find:}$$

i)  $2R_2 + R_3 \rightarrow R_3$

ii)  $-2R_2 + R_3 \rightarrow R_3$

3. Interchanging two rows

Any two rows of a matrix interchange each other to facilitate the operation. Interchanging rows is designated by “ $\longleftrightarrow$ ”, i.e., “ $R_i \longleftrightarrow R_j$ ”, which means  $R_i$  interchanges with  $R_j$ .

#### Example 2.22

$\begin{pmatrix} \frac{1}{2} & 0 & 4 \\ 3 & -2 & 1 \\ 4 & 2 & 5 \end{pmatrix}$ , What will be the resulting matrix if the first and third rows are interchanged?  
 $R_1 \longleftrightarrow R_3 = ?$

### Example 2.23

$$A = \begin{pmatrix} 2 & 9 \\ -4 & 2 \end{pmatrix}$$

1. What row operation performed on matrix A will result another matrix which will have -2 in place of -4 and what will the new matrix be?

**Note:** Matrices resulting from the two alternative operations may not be the same, but the element in the required position will be the same.

2. What row operation be performed to get “1” in the 1<sup>st</sup> row instead of 2?

### ii) Inverse of Matrix

For matrix A, there can be identified another matrix called multiplicative inverse matrix,  $A^{-1}$  so that the product of A and  $A^{-1}$  in either order results in the identity matrix, I.

$$\text{Hence, } A A^{-1} = I = A^{-1} A$$

Important conditions concerning inverse of matrix:

- For matrix, A to have multiplicative inverse, it must be square matrix, as a result it is also called inverse of a square matrix. If a given matrix is not a square matrix, it is impossible to find the inverse matrix.
- The multiplicative inverse of matrix A, is also a square matrix having the same dimension as A.
- Not every square has an inverse, and such matrix is called singular matrix.

### Example 2.24

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \text{ has no inverse, though it is square matrix.}$$

- If  $A^{-1}$  is inverse of A, then A is inverse of  $A^{-1}$ .
- Inverse of a matrix is unique, if it exists.

As we discussed above, multiplicative inverse is obtained only for square matrices. To determine the inverse of an nxn square matrix A, for which  $A^{-1}$  exists there are certain steps to be followed.

**Step 1.** Form the augmented matrix (A/I), where A is an nxn square matrix and I is an nxn identity matrix.

Augmented matrix is a combination of two or more matrices written as one. The original two matrices are separated by vertical bar (/).

**Note:** The identity matrix is used here because the product of a matrix and its inverse will give us an identity matrix.

**Step 2.** Perform the elementary row operations on the augmented matrix until the augmented matrix is reduced to the form (I/B), where matrix B is the desired inverse matrix,  $A^{-1}$ , of matrix A.

**Note:** If (A/I) is transformed by elementary row operation in to (I/B), then the resulting matrix B is the inverse of matrix A,  $A^{-1}$ . However, if all zeros are obtained in one or more rows to the left of the vertical line, then  $A^{-1}$  does not exist.

### Example 2.25

Find the inverse of the following matrices if it exists

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & -4 \\ 1 & 3 \end{pmatrix}$$

**Note:** the corresponding identity matrix will have the same dimension as original matrix .E.g. If original matrix is 2x2 matrix, we set a 2x2 identity matrix

Finding the solution for linear equations using inverse method involves the following steps.

Step1. Transform linear equation to matrix form, which we discussed earlier.

Step 2. Find the inverse by transforming (A/I) in to the matrix (I/A<sup>-1</sup>)

Step 3. Find the solution by multiplying A<sup>-1</sup> with constant b, to get the solution.

### **.Example 2.26**

**Application:** A person has \$5000 to invest, part at 5% and the rest at 10%. How much he invest at each rate to yield \$400 per year?

### **Activity 2.5**

Answer the following questions in the spaces provided below.

1. Given a matrix  $A = \begin{pmatrix} 2 & 4 \\ 1 & -1 \end{pmatrix}$ , what row operation (s) must be applied to get

i) 1 in the 1<sup>st</sup> row of 1<sup>st</sup> column

ii) 0 the 2<sup>nd</sup> column of the 1st row

2. Find the solution using the inverse method for

$$4x_1 + 3x_2 = 2$$

$$9x_1 + 7x_2 = 3$$

### **Solving System of Linear Equations using Gauss-Jordan Method**

While inverse method of solving system of linear equation involves transforming

(A/I)  $\longrightarrow$  (I/ A<sup>-1</sup>) and multiplying A<sup>-1</sup> by b; the constant vector, **Gauss-Jordan Method** involves transforming (A/I)  $\longrightarrow$  (I/S) where, S is the required solution. Unlike matrix inverse method, (which is defined only for square matrices, nxn), **Gauss-Jordan Method** works for any m by n system regardless of whether m=n or m≠n. Moreover, it involves shorter steps than inverse method because it does not involve multiplication of inverse by constant.

### **Steps**

The steps are similar to that of inverse method except that in Gauss-Jordan Method we write the constant term instead of writing identity matrix.

1. Augment the equations. This is done by writing a given matrix at the left, and the corresponding constant vector, next to it, at the right.
2. Select and carry out row operations that will convert the given matrix, in to the identity matrix, and apply the same operations to the matrix at the right.
3. Continue the operation until the left (given) matrix becomes the identity matrix. Then, the matrix on the right will be the desired solution.

### **Procedures**

- Working one column at a time on augmented matrix and by using appropriate row operations, first get a “1” in the 1<sup>st</sup> column of 1<sup>st</sup> row.
- Then, using this “1” to obtain all zeros in the 1<sup>st</sup> column, i.e., use multiples of the 1<sup>st</sup> row to get zeros in all places below the “1” obtained in 1<sup>st</sup> procedure.
- Repeat the above by getting a “1” in the 2<sup>nd</sup> column of 2<sup>nd</sup> row and obtain zeros in all places below and above this one in the same column
- Do the same for others.

### Example 2.27

Find solution for the following linear equation using Gauss-Jordan method.

$$3X_1 + 4X_2 = 1$$

$$X_1 - 2X_2 = 7$$

### Systems of Linear Equations and their possible Solutions

#### i) Systems of Linear Equations

There are three conditions (systems) of linear equations.

- 1) A system of linear equations where the dimensions are equal ( $n \times n$ )
- 2) A system of linear equations where the number of rows are greater than number of columns ( $m > n$ )
- 3) A system of linear equations where the number of columns are greater than the number of rows ( $n > m$ ).

On the other hand, there are three possible solutions to a system of linear equations; Unique solutions, Infinite Solutions, and no solution.

#### 1) Matrix solution of $n$ by $n$ Linear Systems

This is the condition where dimensions (i.e., the number of rows and the number of columns) are equal. In other words, it involves the condition where the number of equations and variables are equal.

**Note:** To solve an  $n$  by  $n$  system, we start with the matrix  $(A/b)$  and attempt to transform it into the matrix  $(I/S)$ . One of the three things will result:

- An  $n$  by  $n$  matrix together with the unique solution.

A given  $n \times n$  matrix has **unique solution**, when our attempt to transform  $(A/b)$  to  $(I/S)$  results in the identity matrix to the left side and constant number to the right as in the case

$$\left( \begin{array}{cc|c} 1 & 0 & m \\ 0 & 1 & n \end{array} \right)$$

### Example 2.28

$$\text{ii. } \left( \begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 3 \end{array} \right) \quad \text{ii. } \left( \begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

- A row that is all zeros except in the constant column, indicating that there is **no solution**. A given  $n \times n$  matrix has no solution if our attempt to transform  $(A/b)$  to  $(I/S)$  results in a row consisting of all zero elements except the constant vector as in the case of:

$$\left( \begin{array}{cc|c} 1 & 0 & m \\ 0 & 0 & n \end{array} \right)$$

### Example 2.29

$$\left( \begin{array}{ccc|c} 1 & 0 & 0 & 12 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 4 \end{array} \right), \text{ has no solution, because the above matrix is the augmented form of:}$$

$$X_1 + 0X_2 + 0X_3 = 12 \Rightarrow X_1 = 12$$

$$0X_1 + X_2 + 0X_3 = -4 \Rightarrow X_2 = -4$$

$$0X_1 + 0X_2 + 0X_3 = 4 \Rightarrow 0 = 4. \text{ But, } 0 \neq 4. \text{ Since } 0 \neq 4, \text{ it is inconsistent.}$$

- A matrix in a form different from the above two conditions indicating there are **unlimited number of solutions**. Note that for  $n \times n$  system, this case occurs when there is a row with all zeros including the constant column as in the case of:

$$\left( \begin{array}{ccc|c} 1 & 0 & 0 & m \end{array} \right)$$

$$\begin{array}{cccc} 0 & 1 & 0 & n \\ 0 & 0 & 0 & 0 \end{array}$$

### Example 2.30

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

## Illustrative examples

(a)  $2X_1 + 3X_2 = 17$

(b)  $2X_1 + 5X_2 = 15$

(c)  $2X_1 + 5X_2 = 15$

$$X_1 + 2X_2 = 10$$

$$3.2X_1 + 8X_2 = 20$$

$$3.2X_1 + 8X_2 = 24$$

**(2) Matrix Solution where  $m > n$ .**

A matrix of  $m$  by  $n$ , where  $m > n$  involves a system of linear equations where the number of rows are greater than the number of columns. In other words, the number of equations is greater than the number of variables. Hence, there is at least one excess equation to the number of variables.

To solve an  $m$  by  $n$  system of equations with  $m > n$ , we start with the matrix  $(\mathbf{A}/\mathbf{b})$  and attempt to transform it in to the matrix  $(\mathbf{I}/\mathbf{S})$ . One of the three things will result:

- An  $n \times n$  identity matrix above **m-n** bottom row(s) that are all zeros, giving the **unique solution** as in the case of:

$$\left(\begin{array}{cc|c} 1 & 0 & m \\ 0 & 1 & n \\ 0 & 0 & 0 \end{array}\right)$$

### Example 2.32

(a)  $\left\{ \begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 6 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right\}$  nxn identity matrix

m-n rows are all zeros, i.e, 5-3=2 row elements are all zeros.

(b)  $\left( \begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & -2 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{array} \right)$  m-n, i.e., 3-2= 1. The last row has all zeros.

- A row that all zeros except in the constant column, indicating that there is no solution as indicated in the case of:

$$\left( \begin{array}{cc|c} 1 & 0 & m \\ 0 & 1 & n \\ 0 & 0 & k \end{array} \right)$$

### Example 2.33

$$\left( \begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & 2 \\ \mathbf{0} & \mathbf{0} & \mathbf{4} \end{array} \right) \Rightarrow \text{this row consists inconsistencies as } 0 \neq 4. \text{ Hence, no solution.}$$

- A matrix in a form different from the above two cases indicates that there is **unlimited number of solutions**.

### Example 2.34

$$(a) \left( \begin{array}{ccc|c} 1 & 0 & 2 & -4 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$(b) \left( \begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

### Illustrative Example

$$i) 4X_1 + 5X_2 = 30$$

$$3X_1 + 2X_2 = 19$$

$$2X_1 + 5X_2 = 20$$

$$ii) 4X_1 + 5X_2 = 30$$

$$3X_1 + 2X_2 = 19$$

$$2X_1 + 5X_2 = 30$$

### 3) System of Linear equations where $n > m$

This case involves system of linear equations where there are more variables than equations; hence there exists at least one more variable than the number of equations. For such a system we have either no solution

**Note:** There is no unique solution for this system of linear equations.

Our attempt to transform  $(A/b)$  in to  $(I/S)$  in the case where  $m < n$  will result in:

- A row which is all zeros except in the constant column, indicating that there is no solution as in the case of:

$$\left( \begin{array}{ccc|c} 1 & 0 & 0 & m \\ 0 & 0 & 0 & n \end{array} \right)$$

### Example 2.34

$$\left( \begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 3 \end{array} \right)$$

- A matrix in a form different from above condition, indicating that there are unlimited number of solutions.

$$\left( \begin{array}{ccc|c} 1 & 0 & -1/4 & 12 \\ 0 & 1 & 3/4 & 6 \end{array} \right)$$

**Note:** In general, when a reduced system has more variables than equations, and contains no contradictions, the system is dependent and has infinitely many solutions. Here to find the value of one variable, we will depend on another variable's value.

### Illustrative Examples

$$2X_1 + 2X_2 + X_3 = 36$$

$$X_1 + 3X_2 + 2X_3 = 30$$

### Application1

Suppose a company makes liquid products: prime oil, Mid oil, and Last oil, which contains different amounts of additives  $a_1$ ,  $a_2$  and  $a_3$  per gallon.

| Liquid    | Gallons Made | pounds (lbs) of additives per gallon |       |       |
|-----------|--------------|--------------------------------------|-------|-------|
|           |              | $a_1$                                | $a_2$ | $a_3$ |
| Perim oil | $X_1$        | 1                                    | 1     | 1     |
| Mid oil   | $X_2$        | 1                                    | 2     | 1     |
| Last oil  | $X_3$        | 2                                    | 0     | 1     |

If the amount of additives available are 20 lbs, 30 lbs and 20 lbs respectively for  $a_1$ ,  $a_2$  and  $a_3$ . Then, calculate the total number of gallons produced from each liquid to use up the additives?

## Application2

Special food for athlete is developed from two foods X and Y. The new food is to be designed in such a way that it contains exactly 16 ounces of vitamin A, 44 ounces of vitamin B, and 12 ounces of vitamin C. Each pound of food X contains one ounce of VA, five ounce of VB, and one ounce of VC. On the other hand each pound of food Y contains two ounce of VA, one ounce of VB, and one ounce of VC. Find the number of pounds of each food to be used in the mixture in order to meet the above requirements?

## Markov Chains

**Markov Chain** is defined as a **sequence of experiment, trials** or observations such that the probabilities for the **next state are completely determined by the present state**. Alternatively, Markov chain is a model which is useful in studying the evolution of certain system over repeated trials or experiments.

### Characteristics of Markov Chains

1. The situation occurs in a chance environment consisting of two or more possible outcomes that occur at the end of a well defined fixed period.
2. Since the situation involves a multi period case, the probabilities of the evolution tend to be transitional probabilities. **Transitional probability (matrix) is a square matrix having no negative entries and the sum of entries in each row equals to 1.**
3. The process is observed after each transition and is governed by transition probabilities.
4. If a transition probabilities of a system from one period to another remains constant and the outcomes of the system in a particular time period depends only on the previous time, the Markov chain process is called **Homogeneous Markov Chain**.

### Important Assumptions

- The system has finite number of states
- The system condition or state in any given period depends only on its state in the preceding period and transition probabilities.
- Transition matrix (probability) is constant over time and the process of keeping these transition probabilities is called stationary Markov chain.
- Changes in the system may occur once and only once each period.

Dear students, this concept will be more clear when we take illustrative example. Therefore, go through the example carefully.

### Example 2.36

Suppose that a restaurant chain notes that 30 % of the dinners it sells each week are beef dinners, and 70% are other dinners. But the chain manager wanted to increase the proportion of beef dinners and carried out promotional campaign and collected the following information.

|          | Next week |        |
|----------|-----------|--------|
|          | Beef      | Others |
| One week |           |        |
| Beef     | 0.8       | 0.2    |
| Others   | 0.6       | 0.4    |

**Note:** The above matrix of proportion is called transition matrix (probability) and it is a square matrix. Moreover, the current state of affairs or initial state of the system (i.e., 30% and 70%) is called State vector and are written to the left of transition matrix. It is a row matrix having no negative entries for a given transition matrix.

## Interpretation

The number 0.8 means that 80% of those buying beef dinner one week buy beef again the next week. The remaining 20% shift to buying other dinners the next week.

Interpret numbers 0.6 and 0.4 in the above transition matrix?

As we discussed earlier, state vector is written to the left of transition matrix. If we wish to find the proportion that will buy beef or others after one week, we put state vector and transition probability as:

$$\begin{array}{cc}
 \text{Beef} & \text{others} \\
 \left[ \begin{array}{cc} 0.3 & 0.7 \end{array} \right] & \begin{array}{cc} \text{Beef} & \text{others} \\ \left[ \begin{array}{cc} 0.8 & 0.2 \\ 0.6 & 0.4 \end{array} \right] \end{array} \\
 \uparrow & \uparrow \\
 \text{State vector} & \text{Transition matrix}
 \end{array}$$

## Interpretation

After one week transition, 80% of the 30% who bought beef one week will buy it again in the next week and additional 60% of the 70% who bought other dinners one week will shift to beef the next week. Note that those who buy beef the next week come from two sources: from previous buyers of beef itself (80% of the 30%) and from those shifted from those shifted from other dinners (60% of the 70%).

The proportion which will buy beef after one week will be:

$$\begin{aligned}
 &= 0.8(0.3) + 0.6(0.7) \\
 &= \mathbf{0.66}
 \end{aligned}$$

In the same way, the proportion buying other dinners next week will be:

$$\begin{aligned}
 &= 0.3(0.2) + 0.4(0.7) \\
 &= \mathbf{0.34}
 \end{aligned}$$

$$\text{In short, } \left[ \begin{array}{cc} 0.3 & 0.7 \end{array} \right] \left[ \begin{array}{cc} 0.8 & 0.2 \\ 0.6 & 0.4 \end{array} \right] = \left[ \begin{array}{cc} \mathbf{0.66} & \mathbf{0.34} \end{array} \right]$$

Note:  $\left[ \begin{array}{cc} \mathbf{0.66} & \mathbf{0.34} \end{array} \right]$  is used as state vector for week #1 and pre multiplies the transition matrix (which is constant) to obtain state vector of week #2, i.e., state vector of week#2

$$\begin{array}{cc}
 \text{Beef} & \text{others} \\
 = \left[ \begin{array}{cc} 0.66 & 0.34 \end{array} \right] & \begin{array}{cc} \text{Beef} & \text{others} \\ \left[ \begin{array}{cc} 0.8 & 0.2 \\ 0.6 & 0.4 \end{array} \right] \end{array}
 \end{array}$$

$$\begin{aligned}
 \text{The proportion buying beef in week \#2} &= 0.8(0.66) + 0.6(0.34) \\
 &= \mathbf{0.732}
 \end{aligned}$$

$$\begin{aligned}
 \text{The proportion buying other dinner in week \#2} &= 0.2(0.66) + 0.4(0.34) \\
 &= \mathbf{0.268}
 \end{aligned}$$

$$\text{State vector of week\#2} = \left[ \begin{array}{cc} 0.732 & 0.268 \end{array} \right]$$

If we continue this procedure forming a chain, the successive state vector can be shown as:

| Week      | Beef   | others |
|-----------|--------|--------|
| Beginning | 0.3    | 0.7    |
| #1        | 0.66   | 0.34   |
| 2         | 0.732  | 0.268  |
| 3         | 0.7464 | 0.2536 |

4

0.74928

0.25072

•  
•

**Note:** the sum of each state vector must be 1 since they are proportions of a whole. Like wise, rows of any transition matrix must each sum to 1. Note again that the state vector appears to be approaching (0.75 0.25). If we multiply transition matrix by this state vector, the result will be the same state (0.75 0.25) and we call this vector the *Steady vector*, which is of the long term phenomenon.

Dear students, instead of going through all the above calculations to reach the steady state, we can use a short way. This can be done as:

Let  $\begin{bmatrix} V_1 & V_2 \end{bmatrix}$  multiplied with the transition matrix, it will give the **steady vector** itself. Hence, to find this steady vector,

$$\begin{bmatrix} V_1 & V_2 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix} = \begin{bmatrix} V_1 & V_2 \end{bmatrix} \text{ but } V_1 + V_2 = 1$$

The matrix multiplication yields:

$$0.8 V_1 + 0.6 V_2 = V_1 \Rightarrow 0.8 V_1 - V_1 + V_2 = 0 \Rightarrow -0.2 V_1 + V_2 = 0 \dots\dots\dots \text{equation 1}$$

$$0.2 V_1 + 0.4 V_2 = V_2 \Rightarrow 0.2 V_1 + 0.4 V_2 - V_2 = 0 \Rightarrow 0.2 V_1 - 0.6 V_2 = 0 \dots\dots\dots \text{equation 2}$$

**Note:** Equation 1 and 2 are one and the same (e.g. if we multiply equation 1 by -1, it will give us equation 2). Therefore, we will take **one of the equations**. On the other, the sum of state vector is one. Hence,

$$V_1 + V_2 = 1 \dots\dots\dots \text{equation 3.}$$

$$\therefore -0.2 V_1 + V_2 = 0 \dots\dots\dots \text{equation 1} \Rightarrow -2 V_1 + 6 V_2 = 0 \dots\dots \text{multiply both sides by 10}$$

$$V_1 + V_2 = 1 \dots\dots\dots \text{equation 3.}$$

Now we can solve this system of linear equations using Gauss- Jordan method or solve simultaneously.

$$\begin{aligned} & \left( \begin{array}{cc|c} -2 & 6 & 0 \\ 1 & 1 & 1 \end{array} \right) R_1 \leftrightarrow R_2 \\ & \sim \left( \begin{array}{cc|c} 1 & 1 & 1 \\ -2 & 6 & 0 \end{array} \right) R_2 + 2 R_1 \rightarrow R_2 \\ & \sim \left( \begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 8 & 2 \end{array} \right) 1/8 R_2 \rightarrow R_2 \\ & \sim \left( \begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 1/4 \end{array} \right) R_1 - R_2 \rightarrow R_1 \\ & \sim \left( \begin{array}{cc|c} 1 & 0 & 3/4 \\ 0 & 1 & 1/4 \end{array} \right) \end{aligned}$$

The steady state therefore will be  $(\frac{3}{4} \quad \frac{1}{4}) \Rightarrow (0.75 \quad 0.25)$ .

### Interpretation

In the long run, the proportion of those buying beef will be 0.75 and that of other dinners will be 0.25.

### Absorbing Markov Chains

When one state totally absorbs or dominates all the other states, the chain leading to this steady state is called **Absorbing Markov Chain**. Consider the example of beef and other dinners. Suppose that the customer is satisfied in beef dinner and continues to buy it, i.e., no customer shifts to other dinners. This results in a 1 in the upper left corner and zero in the upper right corner of transition matrix. Thus,

$$\begin{array}{cc} \text{Beef} & \text{Others} \\ \text{[ 0.7} & 0.3 ] \end{array} \quad \begin{array}{cc} \text{Beef} & \text{Others} \\ \text{Beef} \left( \begin{array}{cc} 1 & 0 \\ 0.6 & 0.4 \end{array} \right) \\ \text{Others} \end{array}$$

Since majority (60%) of those buying other dinners change to beef, then continue to buy beef, it is reasonable to expect that ultimately all customers will buy beef and the steady state vector will be (1 0). Hence, beef has absorbed all the business, and the chain leading to **Absorbing Markov Chain**.

$$\therefore \text{Steady State} = (V_1 \ V_2) \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = (V_1 \ V_2)$$

$$\left. \begin{array}{l} 1V_1 + 1V_2 = V_1 \dots \text{Equation 1} \\ 0V_1 + 0V_2 = V_2 \dots \text{Equation 2} \end{array} \right\} \text{ take either of equations.}$$

$$V_1 + V_2 = 1 \text{ Equation 3}$$

Since  $V_2 = 0$  (from equation 2), we can substitute this value to either of equations to get the value of  $V_1$ . Hence,

$$\begin{aligned} V_1 + V_2 &= 1 \text{ Equation} \\ \Rightarrow V_1 &= 1 - V_2 \\ \Rightarrow V_1 &= 1 - 0 \\ \Rightarrow V_1 &= 1 \end{aligned}$$

Hence, the absorbing state will be (1 0). This shows that in the long run all the customers but beef dinner.